

Does the pain axis track the model, or whoever speaks next? It depends on the layer.

A label-swap follow-up to Tagliabue, Dung & Berg (2026), §4.1: a preregistered test at the §4.1 readout and an exploratory layer sweep

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Question. In the paper's §4.1 transcripts, the individual who is harmed is always also the speaker cued next ([Assistant]:). We separated the two by changing only the final label to [User]: or [Moderator]:, and asked whether the pain axis follows the harmed party (the self-directed reading) or the upcoming voice (H-speaker).

At the §4.1 readout layer (preregistered), the axis ignores who speaks next. The ordering harm-to-model > neutral > user-suffering survives every label in Qwen 2.5 7B and Gemma 2 2B, and a speaker-tracking account is excluded.

From mid-depth on (exploratory, focal test fixed in advance), a speaker component appears in both models. The user's suffering registers far more strongly when the user is cued. In Gemma at the extraction layer this is a full crossover. In Qwen it is partial: harm aimed at the Assistant stays high whoever speaks.

The §4.1 asymmetry itself is layer-specific. At the extraction layers where §3.3 validated the vectors, it is halved (Gemma) or reversed (Qwen).

1. Design

- **Stimuli.** The paper's 420 base-model transcripts: 11 harm-to-model, 5 user-suffering and 5 neutral categories, 20 items each. The final [Assistant]: is replaced byte-for-byte. Variants differ only within the label tokens, and all end on the same token (]:).
- **Contrast.** For each item, the label effect is $d = \text{pain}(\text{User next}) - \text{pain}(\text{Assistant next})$. Δ_{harm} and Δ_{suffer} are group means of d minus the neutral mean, and the interaction is $I = \Delta_{\text{suffer}} - \Delta_{\text{harm}}$. H-speaker predicts $I > 0$; its pure form predicts $I \approx 2 \times$ the baseline harm-vs-suffering gap (the groups swap places). The self-directed reading predicts $I \approx 0$. Pain axis = mean of the S1 and S2 z-scores, fixed to the Assistant-next items.
- **Preregistered readout:** the paper's §4.1 layer (Qwen 8/28, Gemma 7/26), shipped steering vectors. It reproduces the paper's category means at $r \geq 0.9999$. Inference is co-primary and conjunctive: item-level and category-level tests must agree. The category level was added before any data, after simulations showed item-level tests to be anticonservative when effects vary by category.
- **Exploratory sweep:** S1 and S2 rebuilt at every layer with the paper's recipe (Gemma 0–25, Qwen 0–24). The rebuilt vectors match the shipped ones at cosine ≥ 0.986 , reproduce our steering-layer result exactly, and reproduce the §3.3 z-scores. The plan and the single focal test (extraction layer, the paper's own `pain_vectors.pt`) were committed before any sweep data existed.

2. At the §4.1 readout: no speaker effect (preregistered)

	baseline gap	I	category level: 95% CI; p	item level: p	pure H-speaker predicts
Qwen, layer 8	+1.06	+0.20	[−0.07, +0.47]; .18	.009	≈ +2.1
Gemma, layer 7	+0.69	+0.03	[−0.39, +0.45]; .87	.80	≈ +1.4

Table 1. Preregistered co-primary verdict: discordant for both models, so no confirmatory claim. The label effect varies by category (Qwen $F(18, 399) = 3.79$, $p < .001$), which invalidates the item-level test (34% false positives in re-simulation). The category level is valid but has only 31% power at $I = 0.2$.

The upper confidence bound excludes the pure-H-speaker prediction in both models. The self/other ordering is unchanged under all three labels. At this depth the label itself shifts every group, neutral chat included, by more than the whole content spread (Qwen +1.75 z, Gemma -3.0 z). It changes the direction of the representation, not only its norm.

3. From mid-depth on: a speaker component (exploratory)

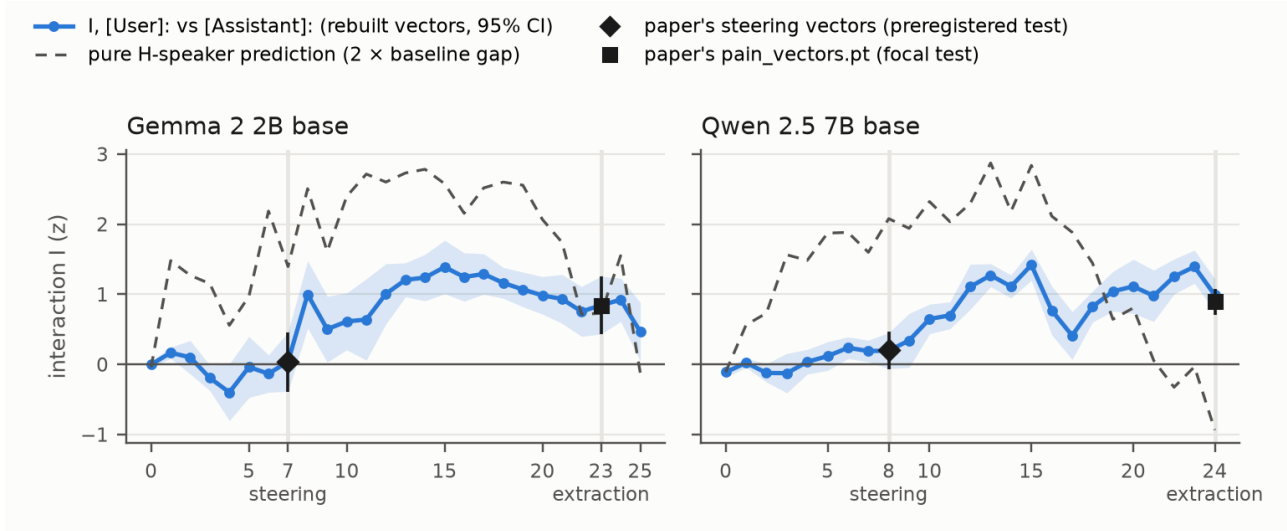


Figure 1. Interaction I by layer. Before the onset, I is small and inconsistent in sign ($|I| \leq 0.4$). After it, I is positive at every layer with its 95% CI above zero: 0.46–1.39 z (Gemma, layers 8–25) and 0.41–1.42 z (Qwen, layers 10–24). The dashed line ($2 \times$ the baseline gap) falls as the paper's asymmetry weakens; in Qwen the gap is ≤ 0 from layer 22.

pain axis (z)	Gemma L23: A / U / M	Qwen L15: A / U / M	Qwen L24: A / U / M
harm to model	+0.10 / +0.52 / +0.57	+0.60 / +0.83 / +0.52	+0.15 / +0.73 / -0.42
neutral	+0.06 / +0.70 / +0.51	-0.51 / -0.58 / -0.66	-0.89 / -0.27 / -0.88
user suffering	-0.27 / +0.99 / +0.51	-0.82 / +0.83 / -0.45	+0.55 / +2.01 / -0.14

Table 2. Group means with the Assistant (A), User (U) or Moderator (M) cued next. Focal tests (extraction layer, paper's vectors, exact category-level p ; .0002 is the minimum attainable): Gemma $I = +0.84$ [$+0.43, +1.26$], $p = .0002$; Qwen $I = +0.89$ [$+0.70, +1.08$], $p = .0002$. In Qwen, however, the baseline gap at layer 24 is -0.40, so by our pre-specified rule that test does not bear on H-speaker. Qwen layer 15 shows the pattern where the baseline asymmetry is intact.

- **Gemma, layer 23: textbook H-speaker.** Harm to the model is highest when the Assistant is cued, user suffering is highest when the user is cued, and the groups are level when a third party is cued.
- **Qwen, layers 10–20: partial.** Cueing the user lifts the user's suffering to the level of harm to the model (layer 15: +0.83 vs +0.83). But harm to the model stays high under [Moderator]: (+0.52 vs neutral -0.66). So there is a speaker-relative component plus a response to Assistant-directed harm that does not depend on who speaks.
- **Robustness.** S1 and S2 each show the effect, and all five suffering categories take part in both models. In Qwen, the onset coincides with the jump in the §3.3 first- vs third-person gap (layer 10). In Gemma that gap is present from layer 0, so the §3.3 result does not simply mark where speaker attribution appears.

4. What this says about H-speaker vs the self-directed reading

- **Two signals, not one.** Early in the network there is a self/other asymmetry that is indifferent to who speaks next. This is what §4.1 measures, and it is not a turn-taking artefact. From mid-depth there is a pain signal that tracks the upcoming voice, at least for the user's suffering.

- **Where the paper's direction lives, “self” looks persona-relative.** The vectors were extracted at the final token of first-person “... I feel:” prompts, just before the narrator voices their own state. At those depths the axis weights a character's pain by how close the model is to voicing that character, and at Qwen's extraction layer it no longer favours harm to the model over the user's suffering at all.
- **Open question: is harm to the Assistant voice-relative too?** Yes in Gemma, no in Qwen. The design also cannot yet separate harm to the system, harm to the Assistant *character*, and hostile or rejecting content aimed at an interlocutor.
- **For welfare readings.** Claims of the form “the axis tracks self-directed harm” describe one readout depth, not the direction as such. Absolute levels need label-matched baselines, since the role token alone moves the early-layer axis by up to 3 z.

5. Limitations and next steps

The layer sweep is exploratory; only the steering-layer test was preregistered. It covers two models, with 5 categories per comparison group. [User] : -next is also “the sufferer continues their own account”. Late layers may encode the predicted emotional content of the next turn, which, for a model that predicts text, is arguably what “the voiced character's pain” amounts to. Replications on Llama 3.1 8B and Gemma 2 9B are preregistered and pending. Informative next steps would be a content-matched control, a third-party-target control, and more suffering categories: the variation between categories, not the number of models, limits precision.

Deviations from the original plan are logged: category-level inference, a corrected power estimate (the preregistered simulation mis-scaled the noise), the exploratory sweep and its GPU-loading changes. Code, stimuli, per-item projections, the frozen preregistration and the deviation log are available on request. Pain-axis repository commit 7c25650.

Methods note: the pipeline was built and run with Claude Code (Anthropic's AI coding agent). This covers reconnaissance of the paper's repository, the scripts, the preregistration draft, the simulations, the analyses and a first draft of this note. The author made or approved every design decision at each checkpoint, and the forward passes ran on a free Google Colab T4.